Travelling Salesman Problem

AI-9.py - C:/Users/Vineela/OneDrive/Documents/AI-9.py (3.15.0)

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```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
min_cost = float("inf")
best_path = None

for route in permutations(range(1, n)):
    path = (0,) + route + (0,)
    cost = 0

    for i in range(len(path) - 1):
        cost += graph[path[i]][path[i + 1]]

    if cost < min_cost:
        min_cost = cost
        best_path = path

print("Shortest path:", best_path)
print("Minimum cost:", min_cost)
```

IDLE Shell 3.15.0

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Vineela/OneDrive/Documents/AI-9.py ==========
Shortest path: (0, 1, 3, 2, 0)
Minimum cost: 80
>>>
```